1 Planning (15 marks)

Defining the problem (3 marks)

P *h* is the independent variable or vary *h*. [1]

P *Q* is the dependent variable or measure *Q* (allow *t*). [1]

P Keep l constant. [1]

Methods of data collection (5 marks)

M Labelled diagram of apparatus: including labelled measuring cylinder/calibrated beaker to receive water. (Measurement may be credited in the text.) [1]

M Vary position of vertical/larger tube. [1]

M Measure *h* and l with a rule/caliper. [1]

M Measure *d* with a travelling microscope or vernier calipers. [1]

M Measure *t* with stopwatch. [1]

Method of analysis (2 marks)

A Plot a graph of *Q* against *h*. [Allow lg *Q* against lg *h*] [1]

A $\eta = \dfrac{2\pi\rho g d^4}{l \times \text{gradient}}$

Must include gradient and η must be subject of formula. [1]

Safety considerations (1 mark)

S Reasoned method to prevent spills, e.g. use tray/sink/cloths on floor. Reasoned method to prevent injury when adjusting metal/glass tubes by wearing protective gloves. Mark may not be scored from the diagram. [1]

Additional detail (4 marks)

D Relevant points might include [4]

1 Repeat experiment for same *h* and average
2 Method to determine the density of water including method to measure mass and volume and equation
3 Take many readings of *d* and average
4 Relationship is valid if straight line passing through origin [if lg-lg graph allow straight line with gradient = 1]
5 Method to check that tube is horizontal
6 Detail on measuring *h* to the centre of the horizontal tube e.g. add radius of tube
7 Keep temperature of water constant

Do not allow vague computer methods.

[Total: 15]

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